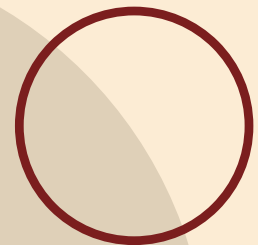
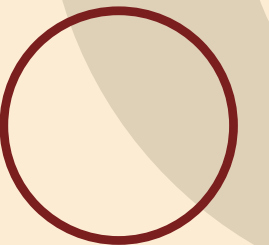
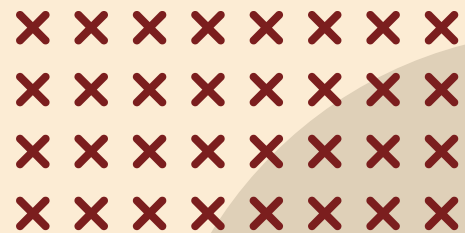


HOUSE PRICE PREDICTION

IT326 project



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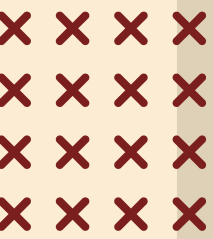
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Problem

Analyzing house prices is challenging due to the many factors that influence them, such as size number of rooms and location These variables make it difficult to identify patterns or accurately predict price categories through manual analysis

Dataset

The dataset used in this project was obtained from Kaggle and contains 6700 records with 14 attributes, including the target variable `quartile_zone`, which classifies each house into one of four price categories. The dataset includes various property related features such as price, number of bedrooms, grade, living area size, real bathrooms, renovation status, basement availability, view quality, condition quality, single floor structure, and time related information such as date and month. This dataset provides a diverse range of characteristics that help analyze property value and support the classification task in this project.



Dataset

date	price	bedrooms	grade	has_basement	living_in_m2	renovated	nice_view	perfect_condition	real_bathrooms	has_lavatory	single_floor	month	quartile_zone
2014-09-26	305000.0	2	1	FALSE	76.18046	FALSE	FALSE	TRUE	1	FALSE	TRUE	9	2
2014-05-14	498000.0	3	2	TRUE	210.88981	FALSE	FALSE	FALSE	2	TRUE	TRUE	5	2
2015-03-23	590000.0	2	4	FALSE	262.91549	FALSE	FALSE	FALSE	2	TRUE	FALSE	3	2
2014-07-15	775000.0	3	3	FALSE	159.79316	FALSE	FALSE	FALSE	1	TRUE	FALSE	7	3
2015-04-14	350000.0	2	1	FALSE	92.903	FALSE	FALSE	FALSE	1	TRUE	TRUE	4	3
2015-05-08	399000.0	2	2	FALSE	201.59951	FALSE	FALSE	FALSE	1	TRUE	TRUE	5	1
2014-10-15	325000.0	1	2	FALSE	94.76106	FALSE	FALSE	FALSE	1	TRUE	FALSE	10	3
2014-07-15	250000.0	3	2	TRUE	177.44473	FALSE	FALSE	FALSE	2	TRUE	TRUE	7	1
2015-02-25	445000.0	3	2	FALSE	360.46364	FALSE	FALSE	FALSE	3	FALSE	FALSE	2	2
2014-07-01	459500.0	1	2	TRUE	116.12875	FALSE	FALSE	FALSE	1	FALSE	TRUE	7	3
2015-03-25	292500.0	2	2	FALSE	194.631785	FALSE	FALSE	FALSE	2	TRUE	FALSE	3	1
2015-04-24	172500.0	1	1	TRUE	140.28353	FALSE	FALSE	FALSE	2	FALSE	TRUE	4	2
2014-05-14	372977.0	2	3	TRUE	157.00607	FALSE	FALSE	FALSE	2	TRUE	FALSE	5	3
2014-08-29	452000.0	1	1	TRUE	113.34166	FALSE	FALSE	FALSE	1	FALSE	TRUE	8	2
2014-10-09	500000.0	2	3	TRUE	160.72219	FALSE	FALSE	FALSE	2	TRUE	TRUE	10	4
2015-01-19	250000.0	3	2	TRUE	203.45757	FALSE	TRUE	FALSE	1	TRUE	TRUE	1	1
2014-08-19	790000.0	3	3	TRUE	208.10272	FALSE	FALSE	TRUE	2	TRUE	TRUE	8	4

Data mining techniques

Data Collection

- What we did: Collected raw dataset from Kaggle “House Price Prediction Treated Dataset”.
- Details: 6700 instances, 14 features (numeric + categorical).
- Class Attribute: `quartile_zone` with 4 labels (1–4).
- Why: To have a structured dataset for classification analysis.
- Outcome: Dataset ready for exploration and preprocessing.

```
In [11]: print("Number of columns:", df.shape[1])
         print("Data types of columns:")
         print(df.dtypes)
```

```
Number of columns: 14
Data types of columns:
date                object
price               int64
bedrooms            int64
grade               int64
has_basement        bool
living_in_m2        float64
renovated            bool
nice_view            bool
perfect_condition    bool
real_bathrooms       int64
has_lavatory         bool
single_floor         bool
month                int64
quartile_zone        int64
dtype: object
```

The dataset has 14 columns (features + class label) , The dataset includes both numeric and categorical attributes

```
In [12]: print("Number of objects (instances):", df.shape[0])
```

Number of objects (instances): 6700

The dataset contains 6700 rows (instances)

```
In [15]: print("Class attribute: quartile_zone")
         print("Unique class labels:", df['quartile_zone'].unique())
         print("Class distribution:")
         print(df['quartile_zone'].value_counts())
```

```
Class attribute: quartile_zone
Unique class labels: [2 3 1 4]
Class distribution:
quartile_zone
2    1954
3    1724
1    1642
4    1380
Name: count, dtype: int64
```

The class attribute is **quartile_zone**.

It has 4 possible values: {1, 2, 3, 4}.

The distribution shows that each label has a sufficient number of instances, which makes the dataset suitable for classification tasks.

Data Understanding & Visualization

Techniques: Histograms, scatter plots, boxplots, bar plots.

Purpose:

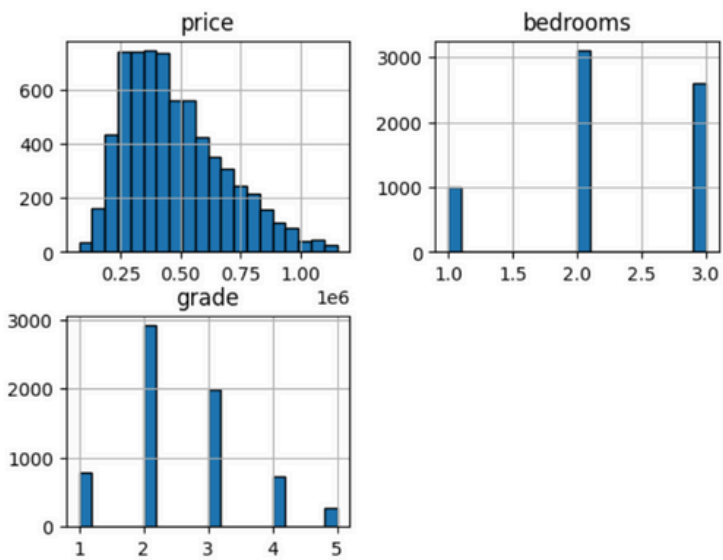
- Understand feature distributions.
- Identify correlations between features and the target (quartile_zone).
- Detect unbalanced categorical features (e.g., renovated, nice view).

Observations:

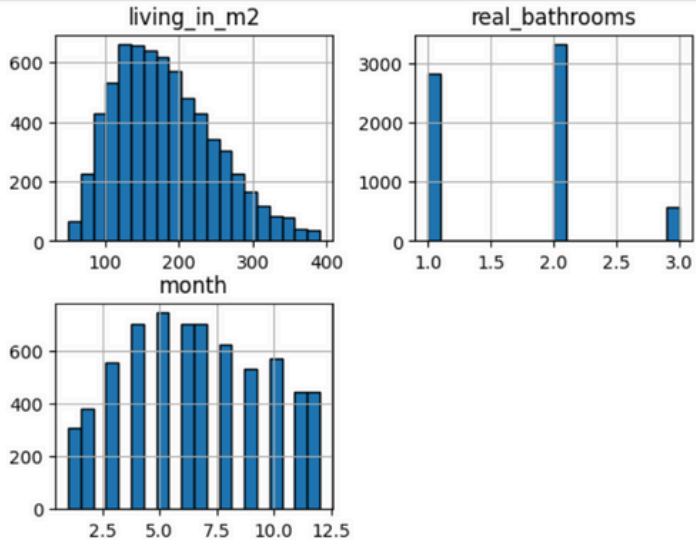
- Price strongly correlated with living area and grade.
- Bedrooms and bathrooms less correlated.
- Some features unbalanced → could bias the model.

Outcome: Insights about data structure and patterns, identified potential preprocessing need

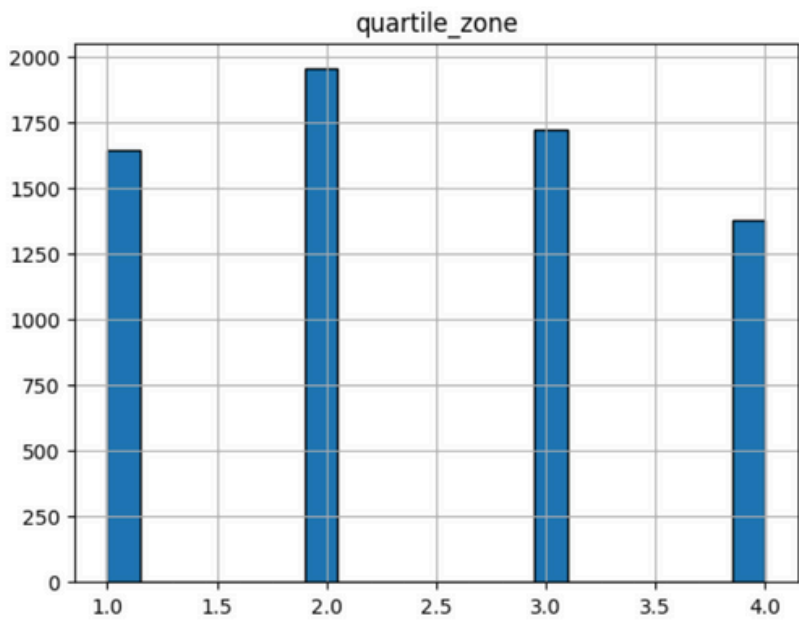
• Histograms



1. Price: The price distribution is right-skewed, meaning most houses are priced between 200,000 and 600,000, with fewer expensive properties above 1,000,000.
2. Bedrooms: Most houses have 2 or 3 bedrooms, while 1-bedroom houses are less common.
3. Grade: Houses are mainly rated with grades 2 and 3, while grades 1 and 5 are much less frequent.

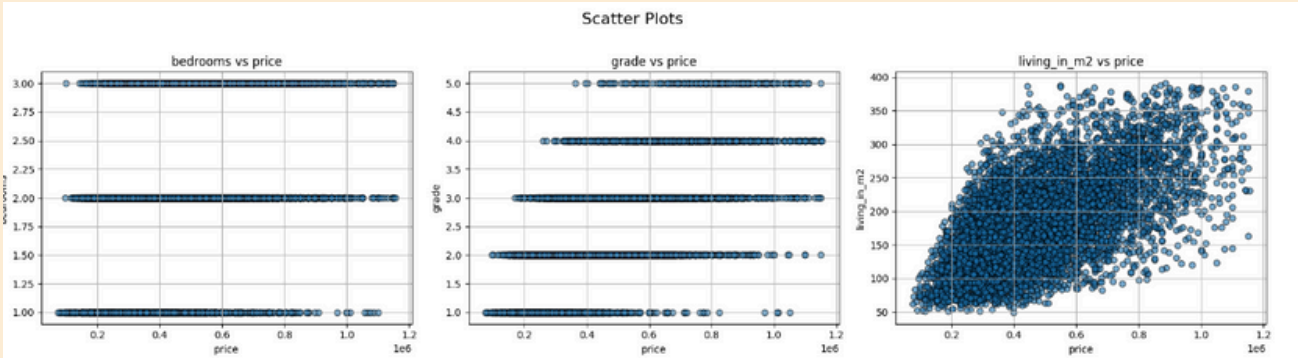


4. Living_in_m2: The living area (in square meters) distribution is right-skewed, with most houses around 100–200 m², and a few larger houses above 300 m².
5. Real_bathrooms: The majority of houses have 1 or 2 bathrooms, with a smaller number having 3.
6. Month : Sales (or recorded data) occur throughout the year, with a slight increase between months 4 and 8, indicating more activity in mid-year



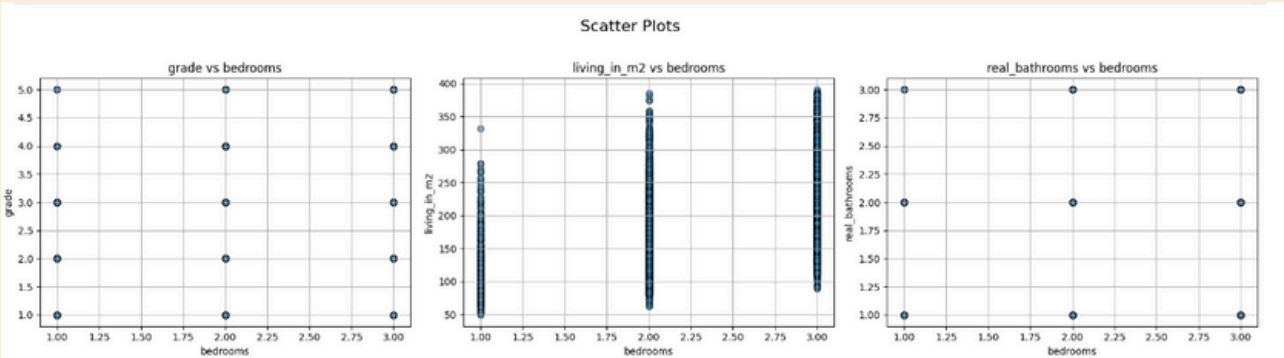
7. Quartile_zone: The data is fairly balanced across the four quartile zones, though Zone 2 has the highest count and Zone 4 the lowest

- Scatter plots



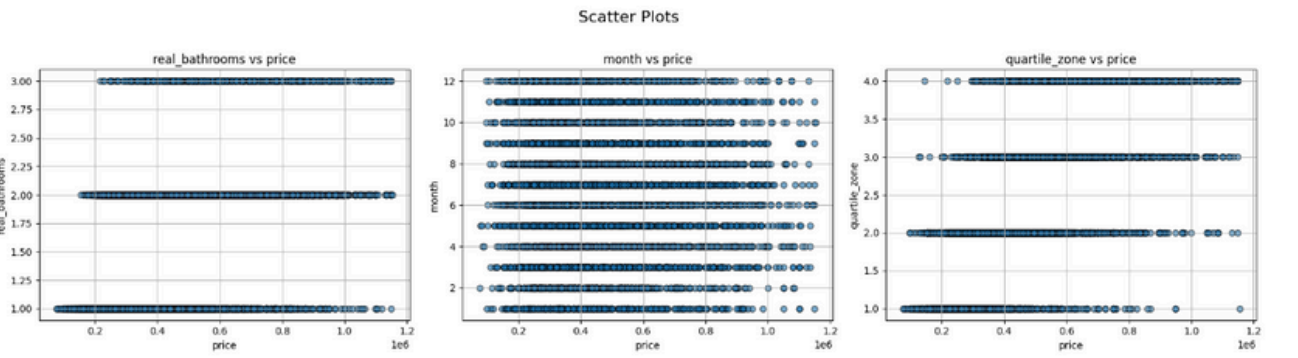
Row1

1. bedrooms vs price: No strong trend—prices vary widely for each bedroom count. Indicates that number of bedrooms alone doesn't explain price differences.
2. grade vs price: Positive correlation—higher grade homes tend to have higher prices, showing quality strongly impacts property value.
3. living_in_m2 vs price: Strong positive relationship—larger houses (in m²) are consistently priced higher.



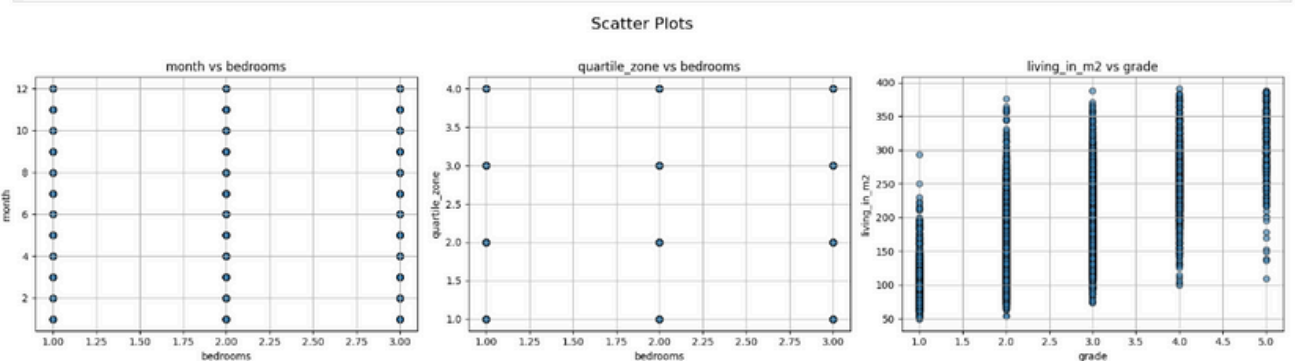
ROW 3

1. grade vs bedrooms: There's a weak relationship; properties with more bedrooms don't necessarily have a higher grade, implying size doesn't directly determine quality rating.
2. living_in_m2 vs bedrooms: A clear positive correlation appears—larger living areas (in m²) tend to have more bedrooms.
3. real_bathrooms vs bedrooms: Slight positive trend; more bedrooms usually come with slightly more bathrooms, though variation is limited.



ROW 2

1. real_bathrooms vs price: The number of bathrooms shows little variation with price; most homes cluster around similar bathroom counts regardless of price, suggesting this variable alone isn't a strong predictor of price.
2. month vs price: Prices are evenly distributed across months, showing no clear seasonal pattern or correlation between month of sale and property price.
3. quartile_zone vs price: Different quartile zones show overlapping price ranges, but higher zones might slightly associate with higher prices, though not strongly.

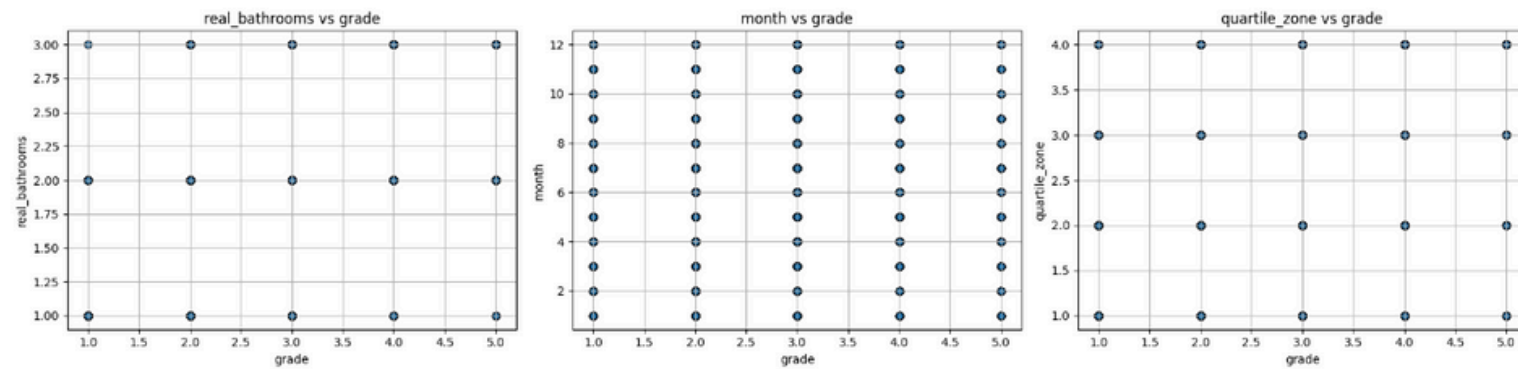


ROW 4

1. month vs bedrooms: Bedrooms are evenly distributed across months—no monthly trend in number of bedrooms sold.
2. quartile_zone vs bedrooms: Bedrooms appear uniformly spread across quartile zones, showing no clear relationship.
3. living_in_m2 vs grade: Positive correlation—higher grade properties generally have larger living areas.

• Scatter plots

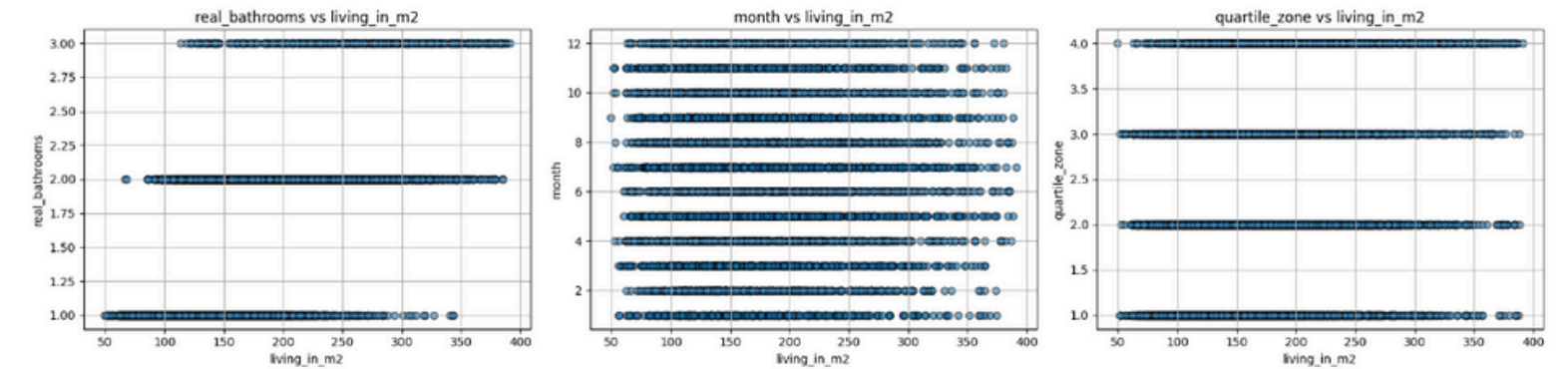
Scatter Plots



Row 5

1. real_bathrooms vs grade: Slight upward trend; homes with more bathrooms tend to have higher grades, though data is limited.
2. month vs grade: No visible pattern; grade doesn't depend on the sale month.
3. quartile_zone vs grade: Grade seems evenly distributed across quartile zones, suggesting minimal connection.

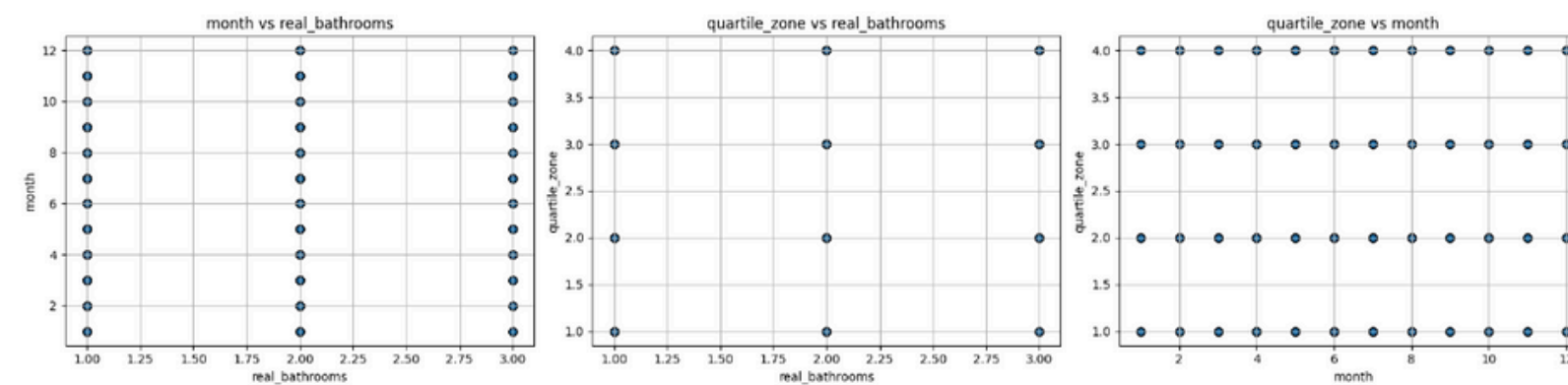
Scatter Plots



ROW 6

1. real_bathrooms vs living_in_m2: Mild positive relationship—homes with more bathrooms tend to have slightly larger living spaces.
2. month vs living_in_m2: No visible trend—property size doesn't vary by sale month.
3. quartile_zone vs living_in_m2: Slight tendency for higher quartile zones to include larger living spaces, but overlap exists

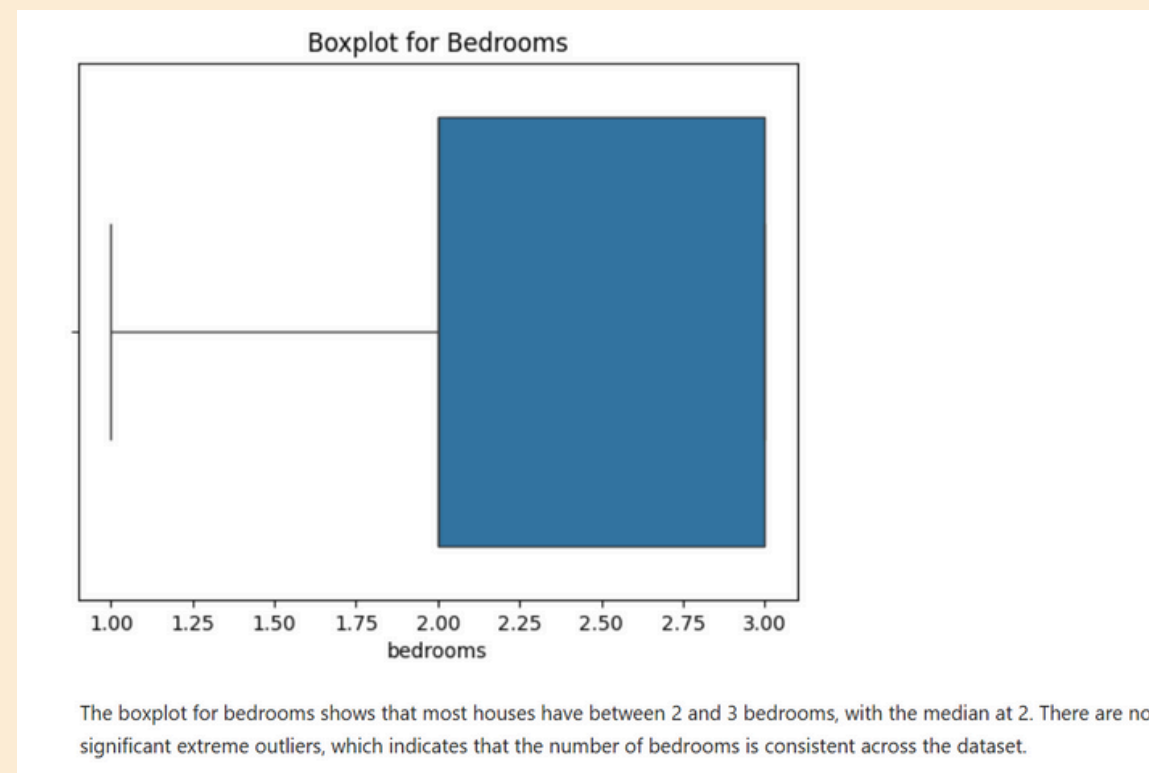
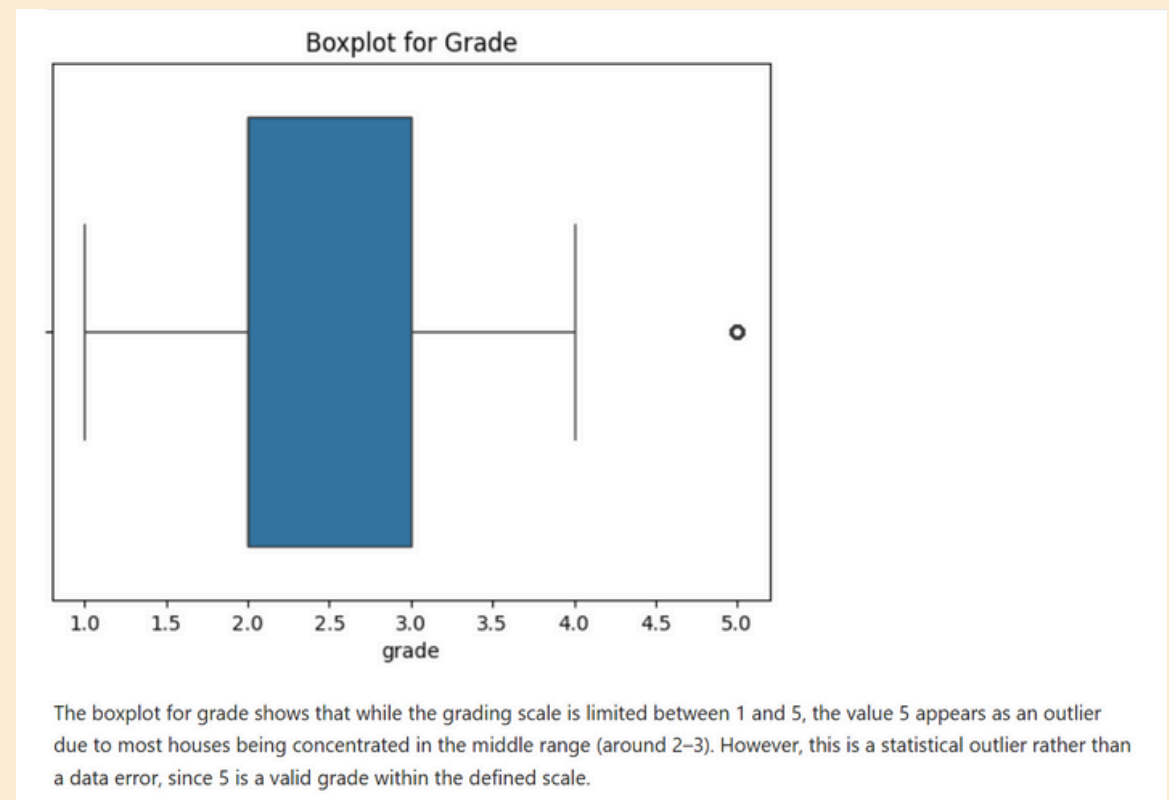
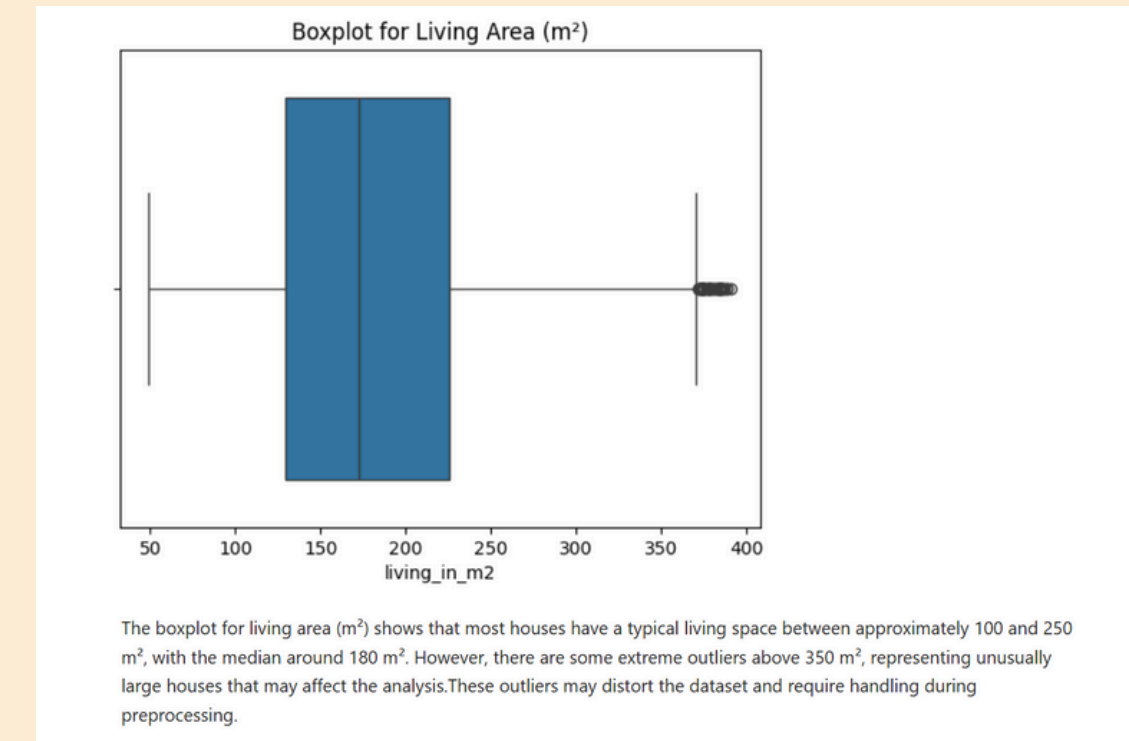
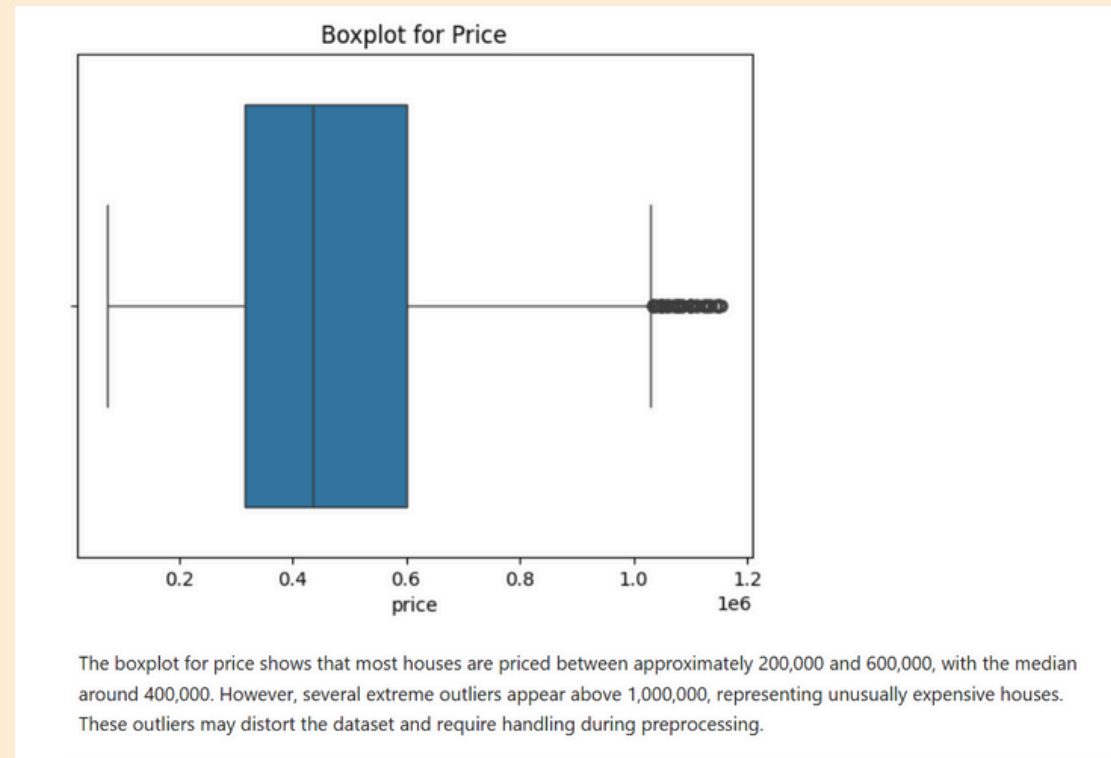
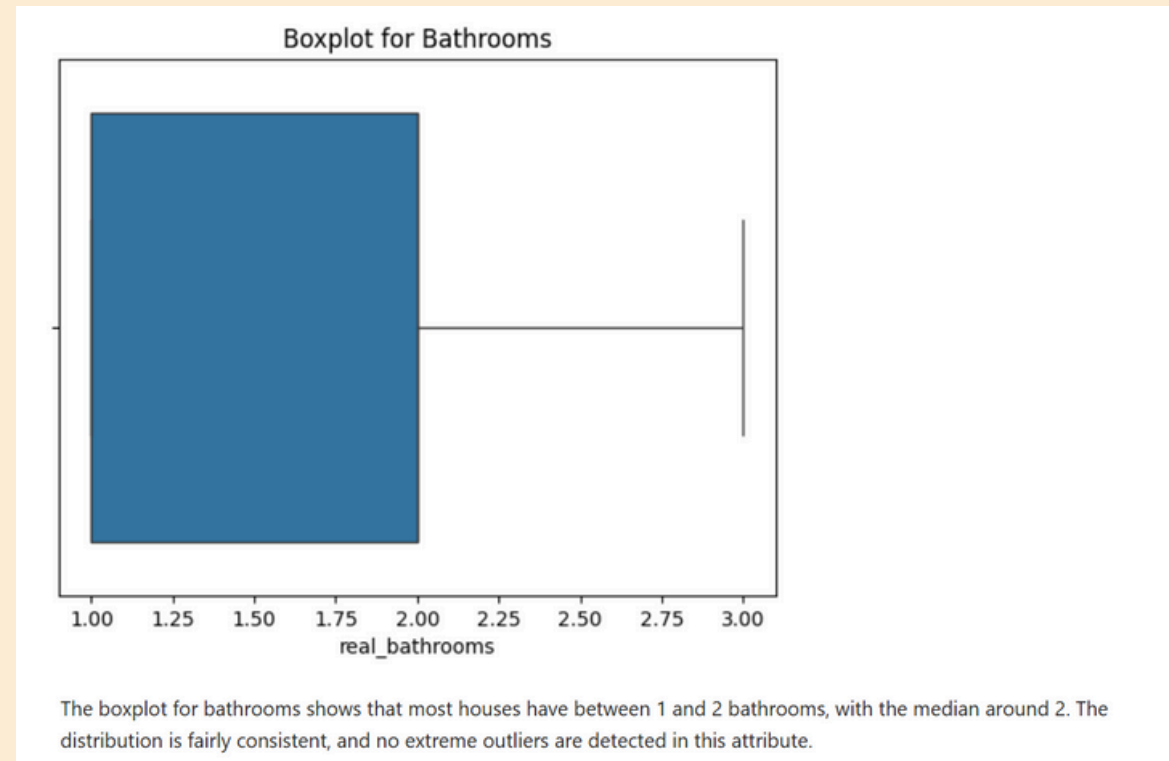
Scatter Plots



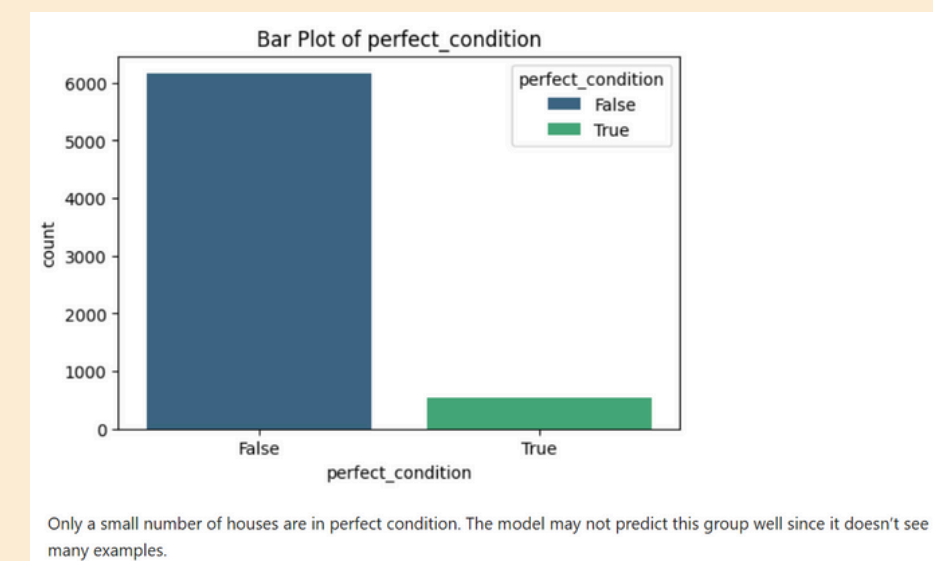
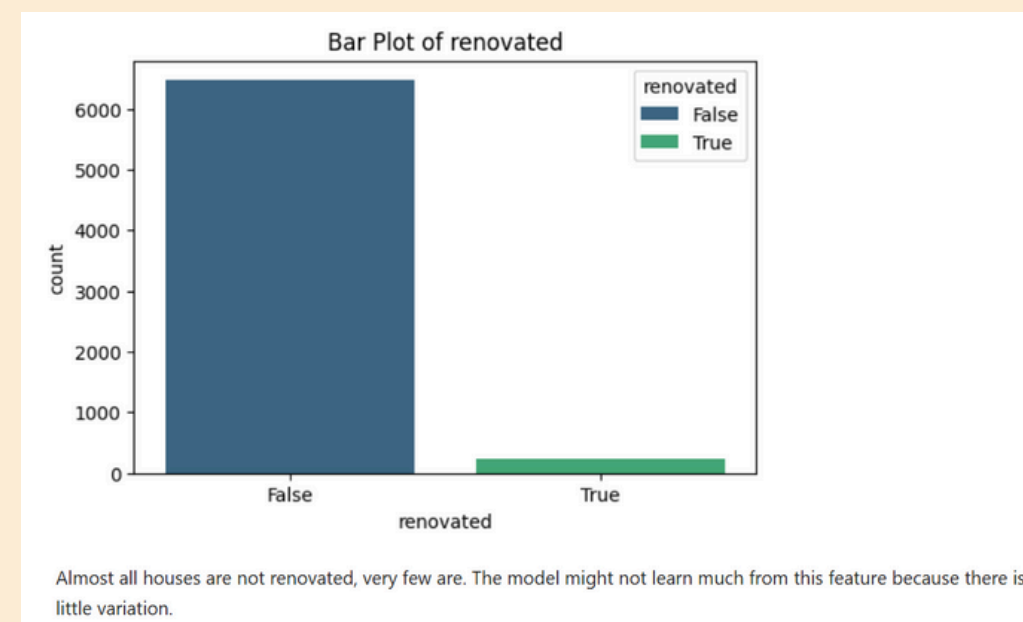
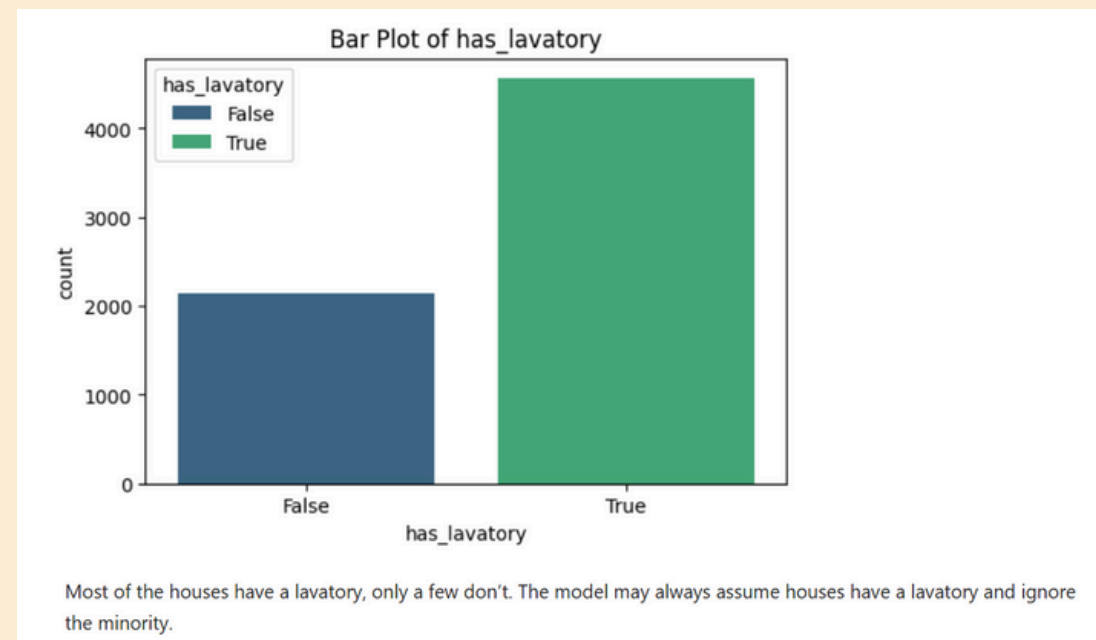
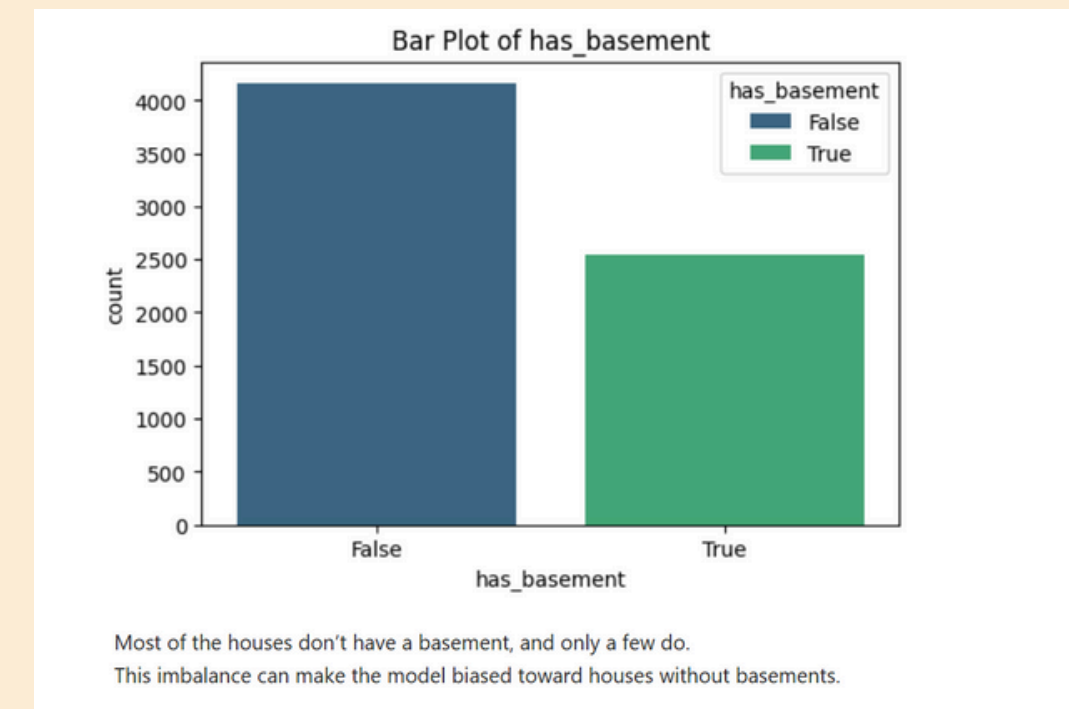
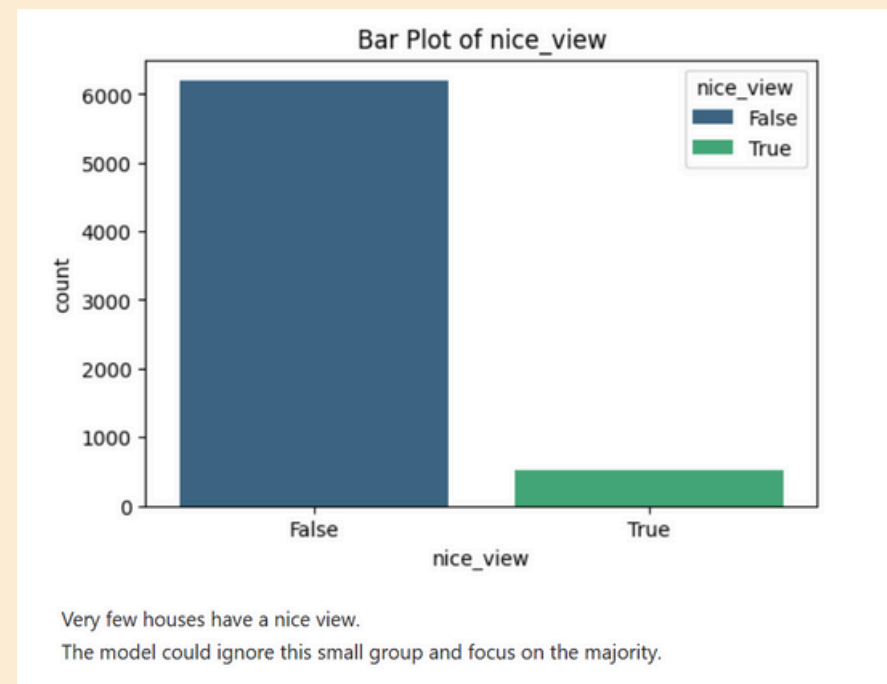
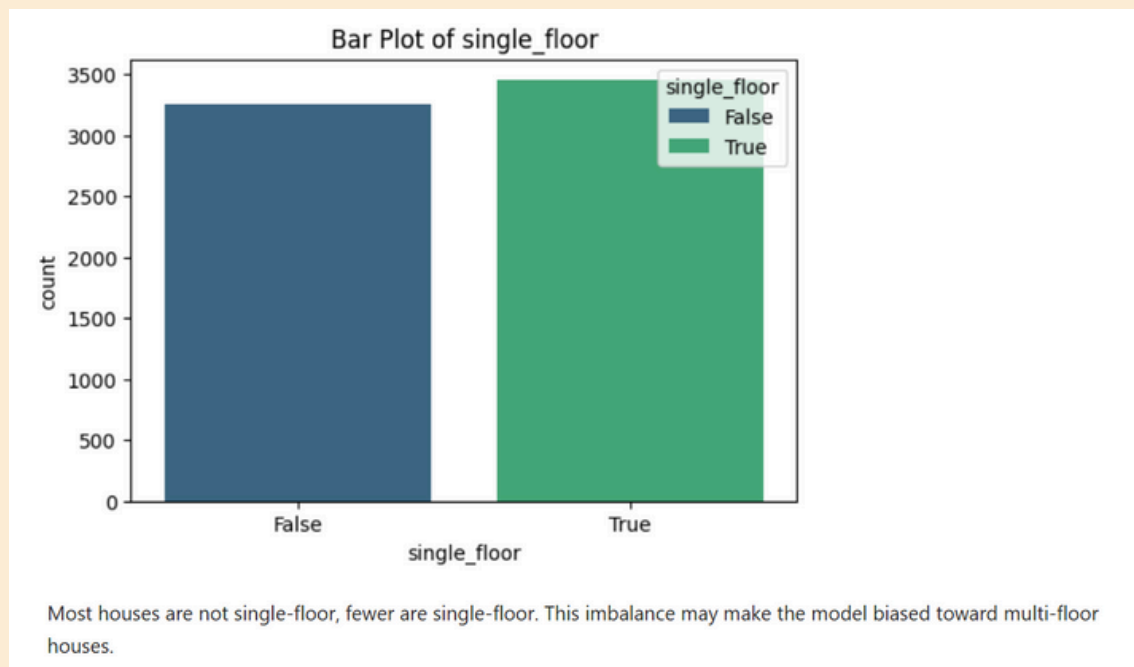
ROW 7

1. month vs real_bathrooms: No trend—number of bathrooms doesn't vary by month.
2. quartile_zone vs real_bathrooms: Bathroom counts appear consistent across quartile zones—no clear pattern.
3. quartile_zone vs month: Quartile zones are evenly represented throughout months—no seasonality effect.

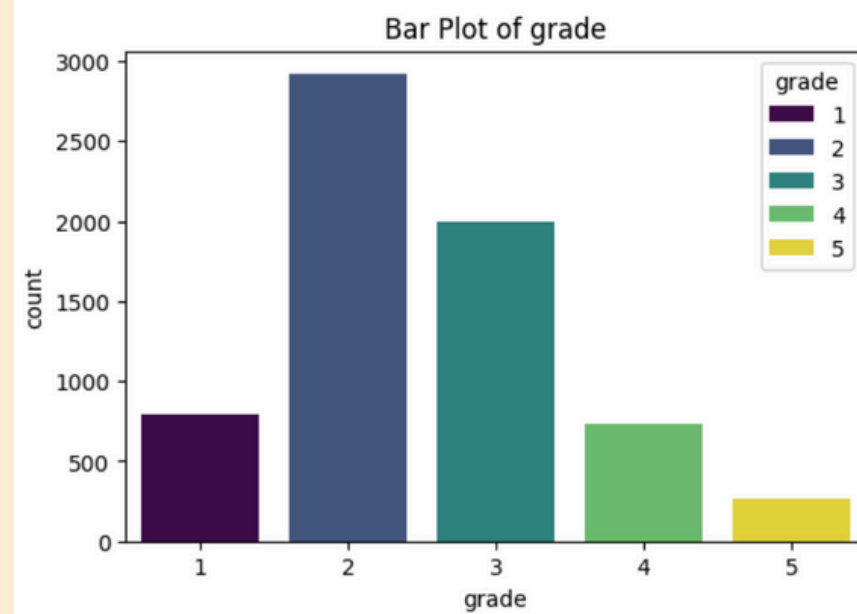
- # Boxplots



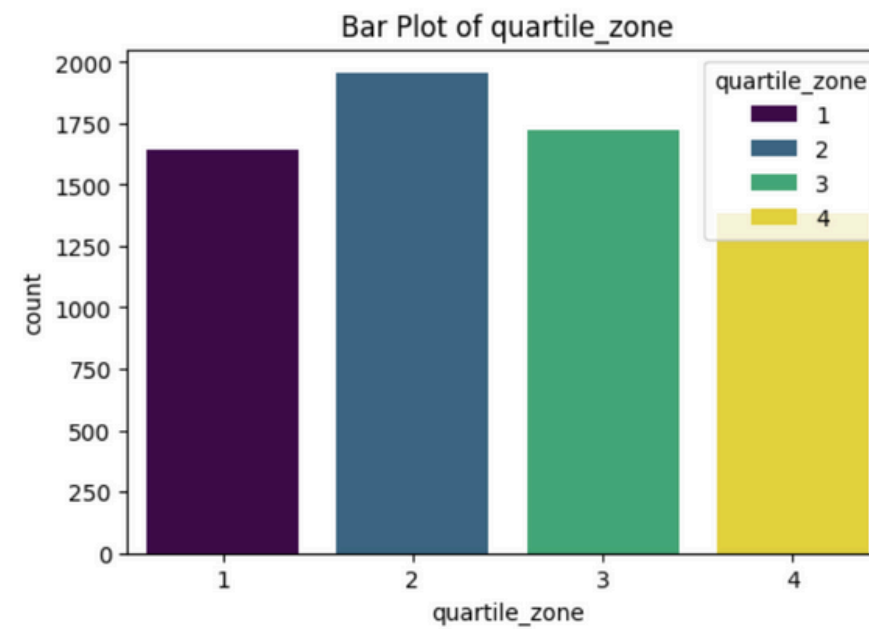
- Bar plots



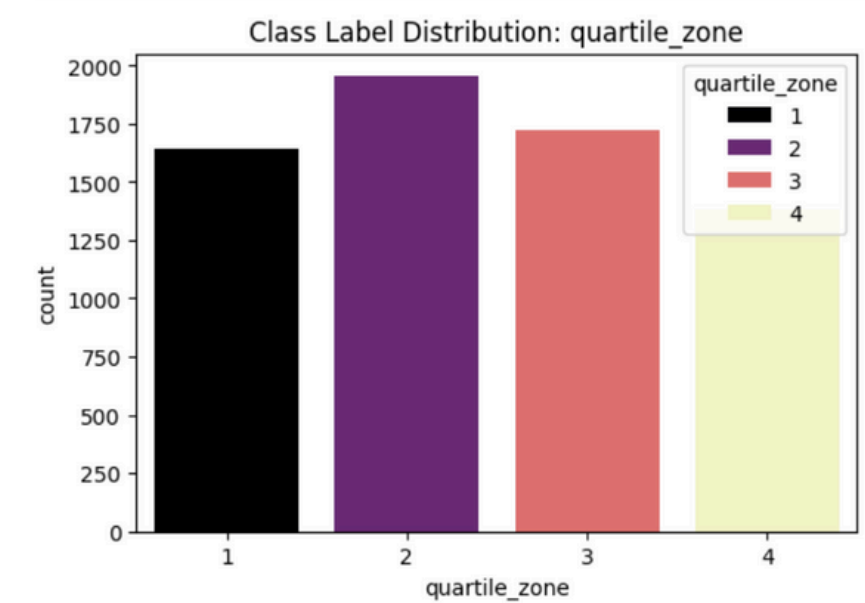
- **Bar plots**



Most houses are in the lower grades (1–3), high grades are rare. The model may struggle to learn about high grades since there are not many examples.



Some zones have more houses than others. The model could perform better on the bigger zones and worse on the smaller ones.



The class labels are not balanced, some zones dominate the dataset. This can cause the model to predict the majority class most of the time.

Standardization

Technique: StandardScaler applied to numeric features (price, living_in_m2, bedrooms, real_bathrooms).

Purpose: Normalize feature scales to prevent bias in the model.

Outcome: Numeric features standardized (mean=0, std=1) → comparable and ready for modeling.

```
In [4]: # Standardization
numeric_features = ['price', 'living_in_m2', 'bedrooms', 'real_bathrooms']

std_scaler = StandardScaler()
df[numeric_features] = std_scaler.fit_transform(df[numeric_features])
```

Standardization was applied because numeric attributes such as price and living_in_m2 had very different scales. Based on statistical summaries from data analysis, these differences could bias the model. Using StandardScaler transformed these columns to a standard scale (mean = 0, std = 1), improving comparability between features.

Feature Selection

Technique: ANOVA F-test (SelectKBest).

Purpose: Identify features most relevant to target variable (quartile_zone).

Process: Evaluated feature scores.

Removed least important: month, bedrooms, real_bathrooms.

Outcome: Dataset noise reduced; only relevant attributes retained → model more efficient

```
In [14]: # Feature Selection - Identify the Least important features

X = df.drop('quartile_zone', axis=1)
y = df['quartile_zone']

X_numeric = X.select_dtypes(include=['int64', 'float64'])

best_features = SelectKBest(score_func=f_classif, k='all')
fit = best_features.fit(X_numeric, y)

feature_scores = pd.DataFrame({
    'Feature': X_numeric.columns,
    'Score': fit.scores_
}).sort_values(by='Score', ascending=True)

least_important = feature_scores['Feature'].head(3).tolist()

df_reduced = df.drop(columns=least_important)
df = df.copy()
df_reduced.to_csv("Preprocessed_dataset.csv", index=False)
```

Feature selection was applied using the ANOVA F-test to identify attributes with the weakest relationship to the target variable (quartile_zone). Based on the statistical scores, the features month, bedrooms, and real_bathrooms showed the lowest importance and were removed. This step helped reduce data noise and improve the overall quality of the dataset by keeping only the most relevant attributes.

Outlier Removal

Content:

- Technique: Interquartile Range (IQR) method on numeric features.
- Purpose: Remove extreme values that distort patterns.
- Outcome:
 - Dataset cleaner, more stable.
 - Reduces the chance of model bias from extreme house

```
In [21]: # Noise Removal
for col in numeric_features:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df = df[(df[col] >= lower) & (df[col] <= upper)]
```

Outlier removal was performed using the Interquartile Range (IQR) method. Boxplot analysis in the previous phase showed extreme values in numeric columns such as price and living_in_m2. Removing these outliers reduced noise and improved dataset stability and reliability.

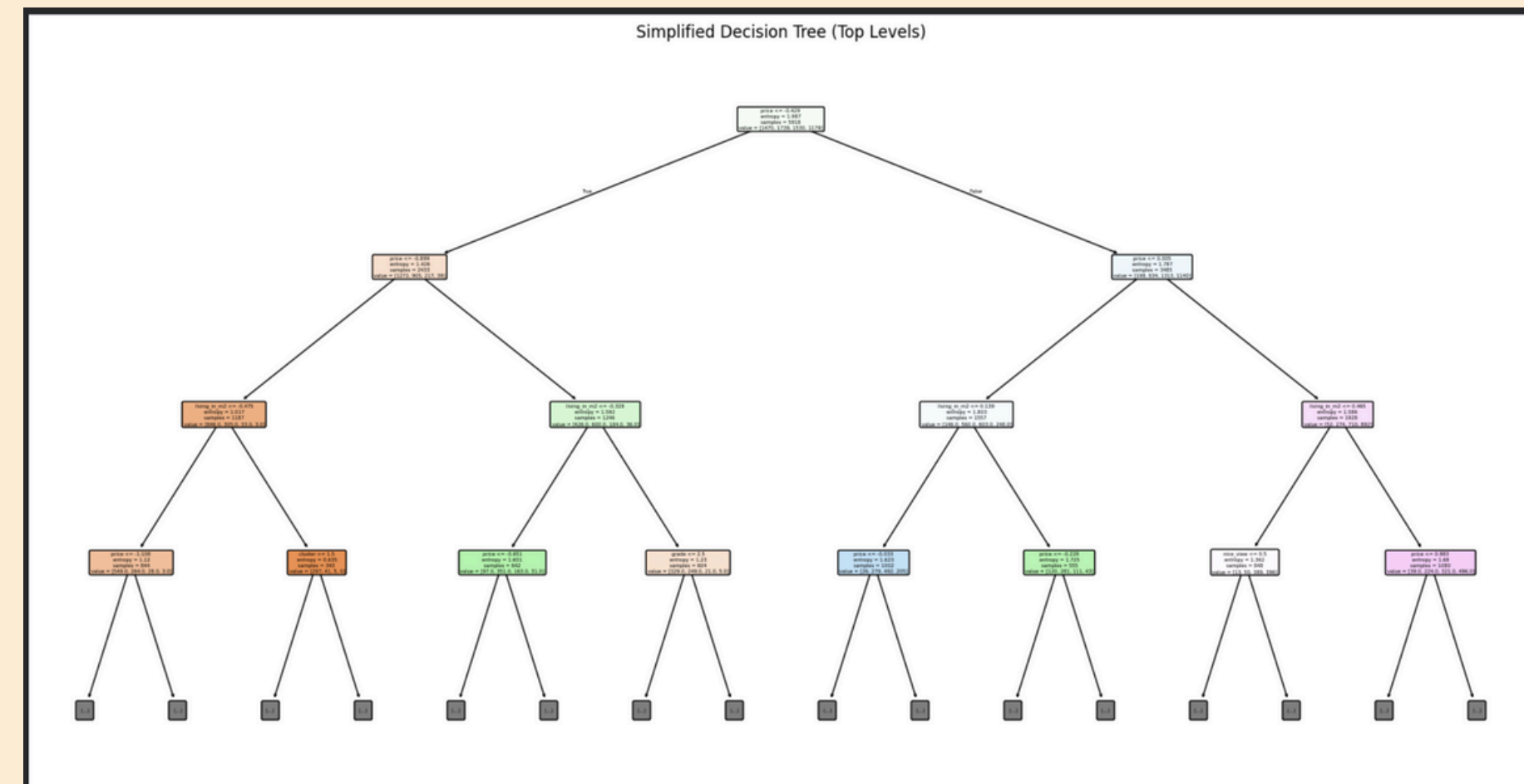
Classification

Technique: Train multiple Decision Tree models using different train/test splits (70/30, 80/20, 90/10) and two criteria (gini, entropy).

Purpose: Identify the best model to classify houses into quartile zones and understand which features drive the classification.

Observations: Larger training sizes improved accuracy.

Outcome: An interpretable Decision Tree summarizing how houses are assigned to each quartile.



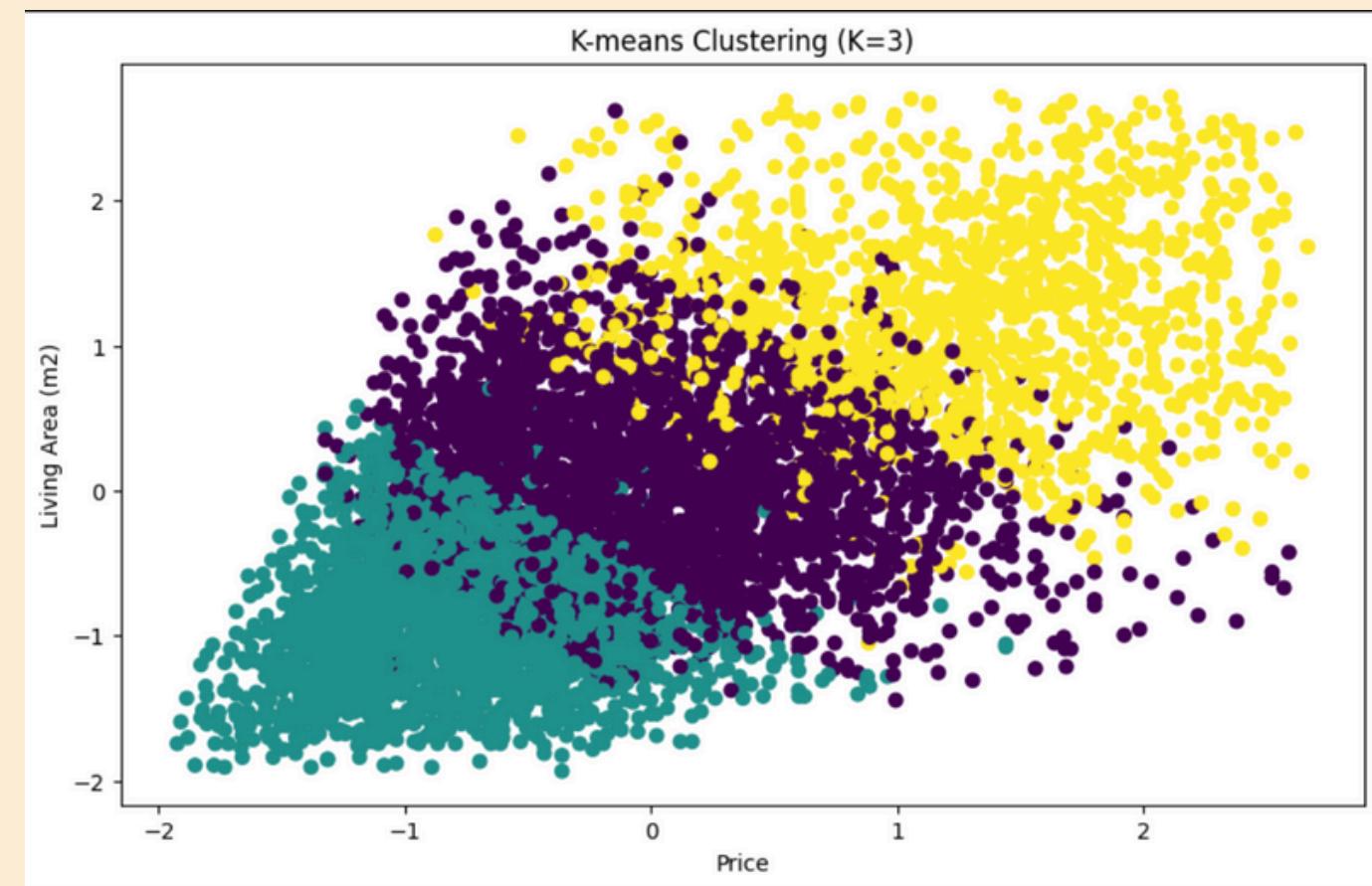
Clustering

Technique: K-Means clustering applied to numeric features, tested with multiple values of k (2, 3, 4) to explore natural grouping patterns.

Purpose: Group similar houses and reveal natural patterns across their numeric characteristics without using labels.

Observations: Different values of k revealed consistent structure. Houses formed clusters based on combinations of price level, living area, and overall grade.

Outcome: Clustering uncovered meaningful groups of similar houses, highlighting variations in property characteristics.



Results

- The dataset was successfully explored, visualized, and preprocessed through scaling, feature selection, and outlier removal, resulting in a structured and stable dataset ready for modeling.
- The Decision Tree achieved the best classification performance using the 90/10 split, accurately predicting the housing price quartile (Q1–Q4).
- K-Means revealed that $K = 2$ was the most meaningful grouping, forming two distinct clusters of houses differing mainly in quality, size, and overall structure.
- Visualizations confirmed strong relationships between overall quality, living area, year built, and garage capacity and the housing price levels.

Findings

- The classification model provided a reliable and interpretable method to predict house price quartiles, showing that structural and quality attributes strongly influence pricing.
- The clustering analysis uncovered natural market segments, highlighting clear separation between high-quality, high-area homes and more moderate ones.
- Together, both techniques offered a complete understanding:
 - Classification → accurate supervised prediction.
 - Clustering → deeper insight into hidden market structure.
- Overall, the Decision Tree was the most effective solution for the prediction task, while K-Means added valuable analytical insights that helped explain housing patterns.

CONCLUSION

In this project we analyzed the housing dataset then applied KMeans clustering to discover patterns in the data

Our results showed clear groups of houses based on price, size, and quality We found that the optimal number of clusters $k=4$ which gave the best separation and meaningful categories of homes

These clusters helped us better understand trends in the housing market and how different features influence property values.

THANK YOU

